

Multiple Choice (10 marks)

1. Insulin deficiency is associated with
 - (a) Reduced lipolysis
 - (b) Increased ketogenesis
 - (c) Reduced gluconeogenesis
 - (d) Reduced proteolysis
2. What is the best method to estimate changes in body fat during weight loss?
 - (a) BMI
 - (b) DXA
 - (c) Four-compartment model
 - (d) MRI
3. From the list of oral microorganisms, which is primarily responsible for the initiation of dental caries?
 - (a) Mutans streptococci
 - (b) bifidobacteria
 - (c) Lactobacilli
 - (d) P. gingivalis
4. Which of the following inborn errors of metabolism gives rise to zinc deficiency?
 - (a) Acrodermatitis enteropathica
 - (b) Wilson's disease
 - (c) Menkes disease
 - (d) Haemochromatosis
5. The gut microbes metabolize L-carnitine into which metabolite that has been shown increase cardiovascular disease?
 - (a) choline
 - (b) betaine
 - (c) Trimethylamine
 - (d) All of the above

True / False (10 marks)

Answer True or False

6. True or False: Molecules of natural polyunsaturated fatty acids in vegetable oils contain 18 carbon atoms with at least two carbon-carbon double bonds in the trans configuration.
7. True or False: Fish can be classified as goitrogens because they contain substances that impact thyroid health.
8. True or False: Vitamin A does not play a role in the synthesis of blood clotting proteins.
9. True or False: Gluten is the main protein found in maize, which is not accurate.
10. True or False: A sedentary 43-year-old woman who weighs 63 kg requires about 1800 kcal per day to maintain her energy levels.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of niacin as a coenzyme in reduction reactions during fatty acid synthesis, and analyze its importance in human health and metabolism.
12. Discuss the factors that contribute to vitamin and mineral deficiencies in older adults in high-income countries, and explain why high nutrient requirements for tissue turnover do not increase this risk.

End of Paper